

Synoptic CB: Manual Redox Probe

2022-2025 Samples

2026-07-08

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##Grab 2022-2025 Processed Data Files

##Clean field notes

```
unique(final_dat1$Notes)
```

```
## [1] NA
## [2] ""
## [3] "UP too dry to go deeper than 10 cm"
## [4] "Probe 6 is jumpy"
## [5] "6th probe was jumpy; did not use for UP"
## [6] "probe 3 must have hit a pocket; it was checked twice"
## [7] "flooded; hard to tell depths"
## [8] "probe 3 mesured twice; must be a pocket"
## [9] "I feel like these Wetland readings should be negative ??"
## [10] "Couldn't get probes more than 10 cm into ground in upland. "
## [11] "Very dry"
## [12] "5 mV measurement was done at 30 cm"
## [13] "About 6 cm standing water on marsh"
## [14] "4|45 pos change"
## [15] "Probes 2 and 5 were at 30 cm"
## [16] "Probe 2 end popped off, replaced in upland from 20 cm measurement"
## [17] "Probe 3 popped off start of wetland measurements - used bare wire"
## [18] "WC very dry"
## [19] "changed batteries in unit; 35 cm reps had bubbles come up when sank to 35 cm"
```

```

final_dat2 <- final_dat1 %>%
  mutate(Notes = case_when(
    str_detect(Notes, "UP too dry") ~ NA,
    str_detect(Notes, "probe 3 mesured twice; must be a pocket") ~ "probe 3 measured twice; must be a p
    str_detect(Notes, "I feel like") ~ NA,
    str_detect(Notes, "Couldn't get probes more than 10 cm into ground") ~ NA,
    str_detect(Notes, "4|45 pos change") ~ "probe 4 measured at 45 cm",
    str_detect(Notes, "5 mV measurement was done at 30 cm") ~ "probe 5 measurement was done at 30 cm",
    str_detect(Notes, "WC very dry") ~ "Very dry",
    Notes == "" ~ NA,
    TRUE ~ Notes
  )) %>%
  mutate(Notes = str_to_sentence(Notes))

unique(final_dat2$Notes)

```

```

## [1] NA
## [2] "Probe 6 is jumpy"
## [3] "6th probe was jumpy; did not use for up"
## [4] "Probe 3 must have hit a pocket; it was checked twice"
## [5] "Flooded; hard to tell depths"
## [6] "Probe 3 measured twice; must be a pocket"
## [7] "Very dry"
## [8] "Probe 5 measurement was done at 30 cm"
## [9] "About 6 cm standing water on marsh"
## [10] "Probe 4 measured at 45 cm"
## [11] "Probes 2 and 5 were at 30 cm"
## [12] "Probe 2 end popped off, replaced in upland from 20 cm measurement"
## [13] "Probe 3 popped off start of wetland measurements - used bare wire"
## [14] "Changed batteries in unit; 35 cm reps had bubbles come up when sank to 35 cm"

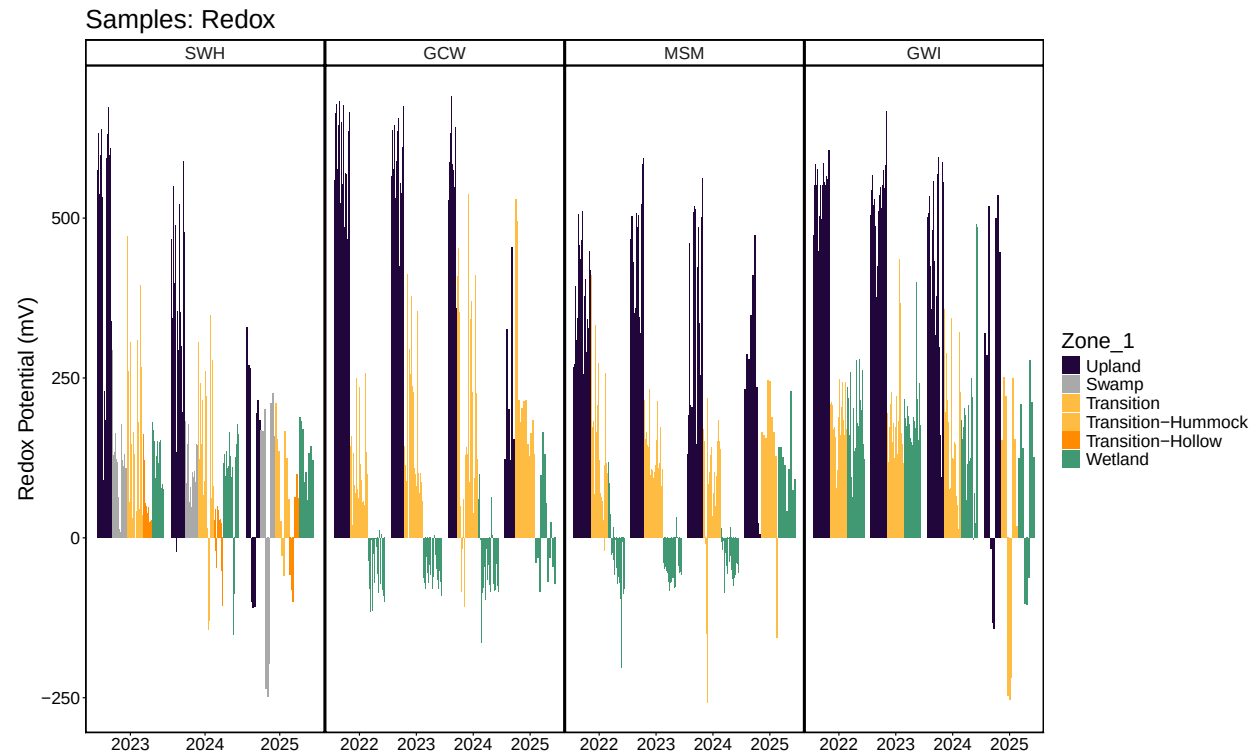
```

```

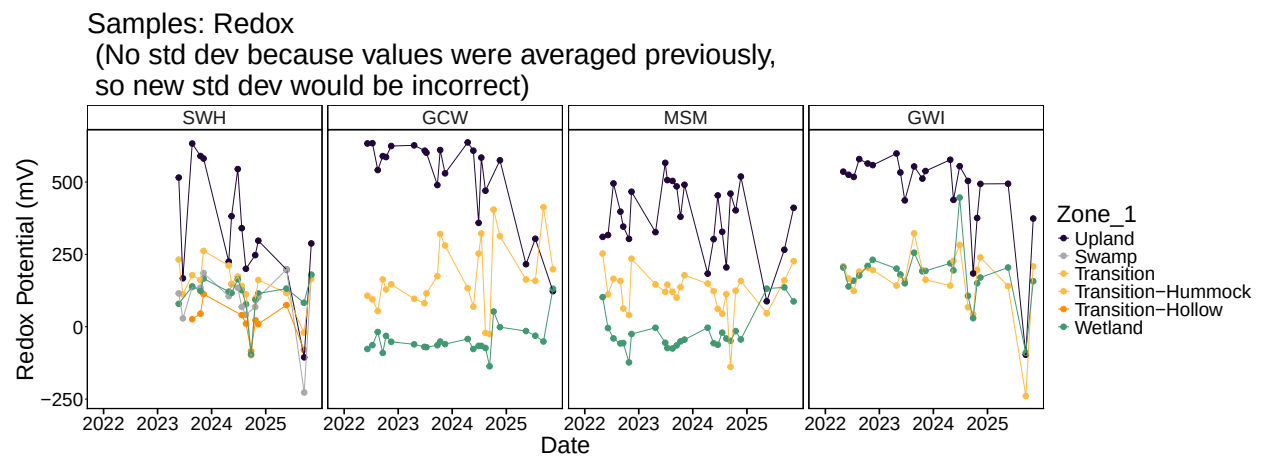
## Write out files

```

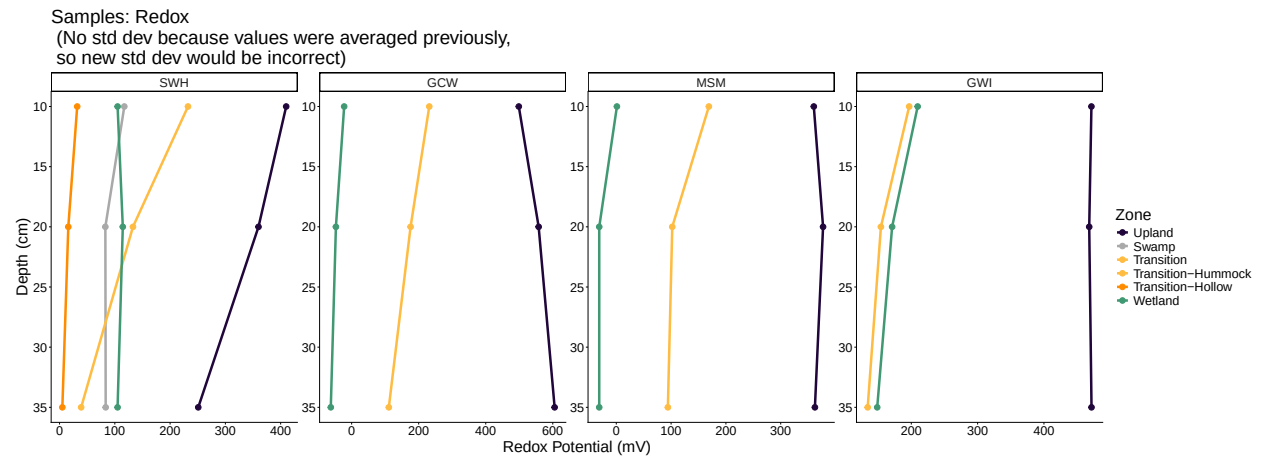
0.1 Visualize Data by Plot



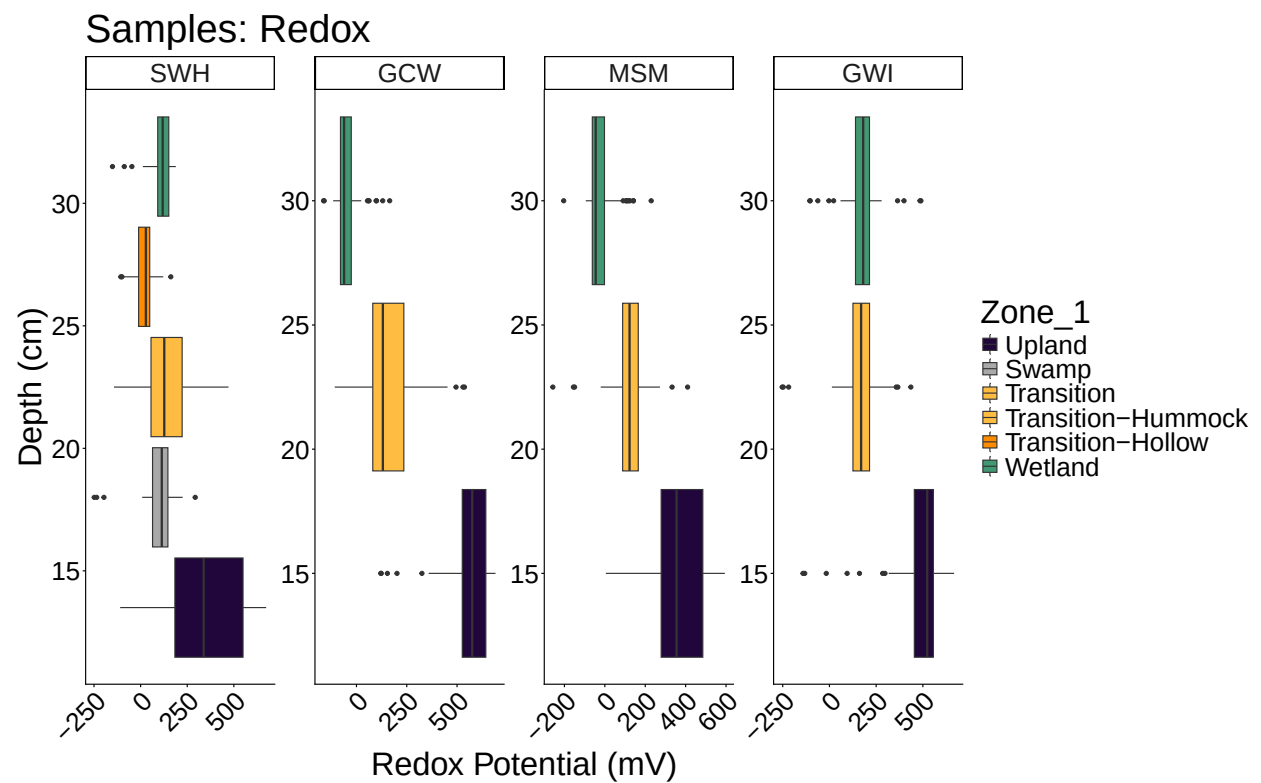
0.2 Summarized data for Site and Zone



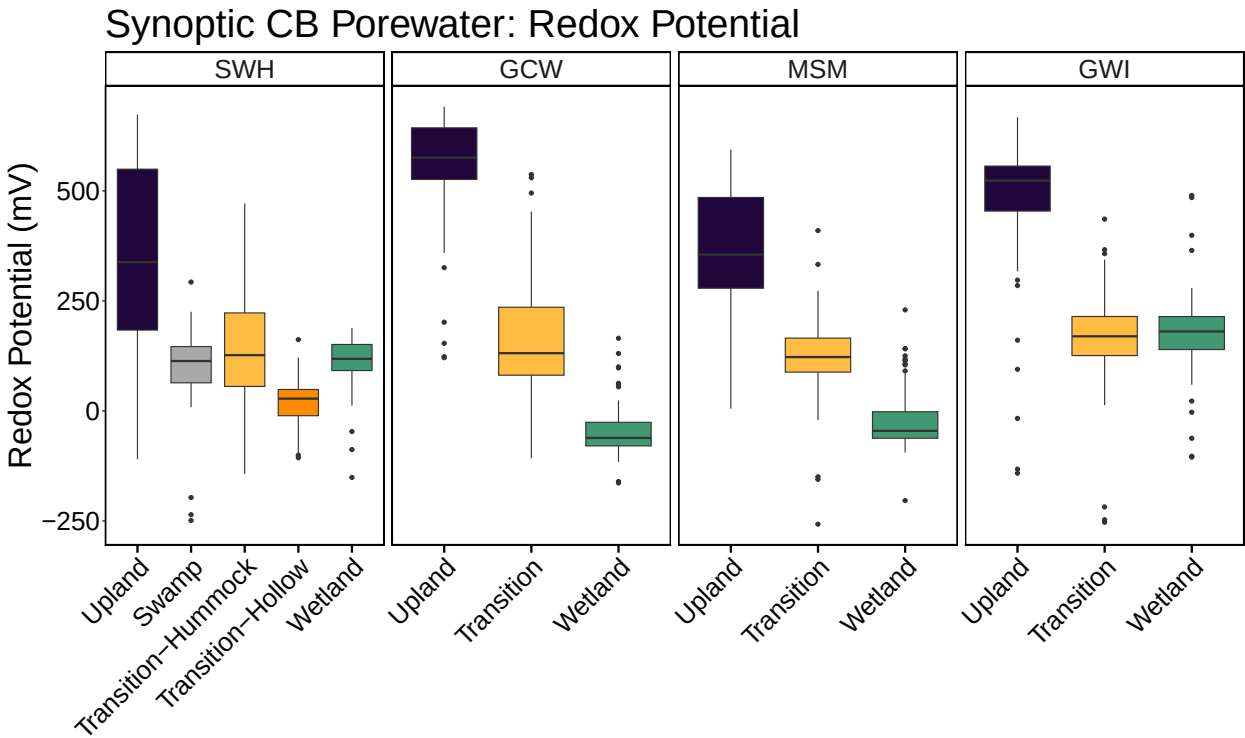
0.3 Summarized Data by Depth



0.4 Boxplot by depth



0.5 Boxplot summary



```
## pdf
## 2

##Calculate summary metrics
##Calculate number of samples
```

0.6 Summary Tables

```
## Warning: package 'kableExtra' was built under R version 4.3.3
```

Table 1: Summary Statistics

Site	Zone_1	min_mV	max_mV	mean_mV
SWH	Upland	-109.8000	673.3833	340.2810
SWH	Swamp	-248.8800	292.8000	94.4988
SWH	Transition-Hummock	-142.9600	471.4000	134.7579
SWH	Transition-Hollow	-106.2400	161.9600	17.6191
SWH	Wetland	-151.0200	188.7800	108.1364
GCW	Upland	121.0000	691.1333	541.5620
GCW	Transition	-107.7167	536.9500	172.7612
GCW	Wetland	-163.5800	165.1800	-43.8218
MSM	Upland	5.0000	593.8667	366.3840
MSM	Transition	-257.2400	409.7200	121.7259
MSM	Wetland	-203.6167	229.5600	-20.3476
GWI	Upland	-141.3800	667.2167	470.5128
GWI	Transition	-252.8600	435.7667	161.7616
GWI	Wetland	-104.8200	489.5800	176.4567

Table 2: Total Samples: 818

Year	Number_of_samples
2022	170
2023	226
2024	296
2025	126

0.7 Number of Samples

Synoptic CB Manual Redox Probe Sample Numbers

